TimeTraveler: Temporal Strategy Planning with Time Dictionary for Streaming Video Understanding

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Abstract

Streaming video understanding (SVU) requires models to answer user queries over continuously evolving video streams. Unlike offline video QA, SVU must handle queries whose evidence may lie in the past, appear in the current scene, or emerge only after future observations. Existing streaming video systems often rely on compact visual memories or fixed evidence-access schemes, which can lose fine-grained details and fail to adapt the answering process to the temporal intent of each query. In this paper, we propose TimeTraveler, a streaming VQA framework that performs temporal strategy planning with a Time Dictionary. Inspired by the need to search across different temporal regions of a stream, TimeTraveler stores each observed moment as a timestamp-indexed structured caption and uses this dictionary as an explicit source of queryable evidence. Given a query, TimeTraveler determines whether to recall past evidence, read the present context, or wait for future observations, and then applies a strategy-specific evidence acquisition process. Comparative experiments on streaming video QA benchmarks show that TimeTraveler improves streaming question answering by preserving fine-grained temporal evidence and selecting the appropriate temporal strategy for each query.

1 Introduction

The rapid progress of Large Vision-Language Models (LVLMs) has expanded the scope of video understanding [1, 2, 3, 4, 5, 6, 7], shifting the field from task-specific recognition toward open-ended visual reasoning [8, 9, 10], temporal localization [11, 12], and interactive question answering [13, 14]. Despite this progress, most video-LLMs [15, 16] are still designed for an offline assumption: a holistic video is given in advance, and the model answers after processing the entire input. This assumption breaks down in streaming scenarios, where visual observations arrive causally, the stream may be unbounded, and user queries can be issued at arbitrary moments. In such settings, the required evidence is structurally heterogeneous. The system must not only support rich perception of incoming visual information, but also preserve past events for later access and remain responsive to events that can only be confirmed through future observation. These requirements make streaming video understanding (SVU) particularly challenging.

Recent studies on SVU fall into two main directions. One develops memory-centric mechanisms [17, 18, 19] that compress and retain stream observations (*e.g.,* prototypes, memory tokens, hierarchical summaries, or compressed KV caches) to preserve past information under limited budgets. The other branch [20, 21, 22, 23] improves online interaction by enabling real-time response scheduling, proactive activation, or streaming adaptation of offline video MLLMs, allowing the system to respond at more appropriate moments as the stream unfolds.

Specifically, memory-based methods preserve evidence by converting the video stream into historical memory [19], caching visual information as KV memory [24, 18], or selectively storing query-relevant

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40th Conference on Neural Information Processing Systems (NeurIPS 2026).

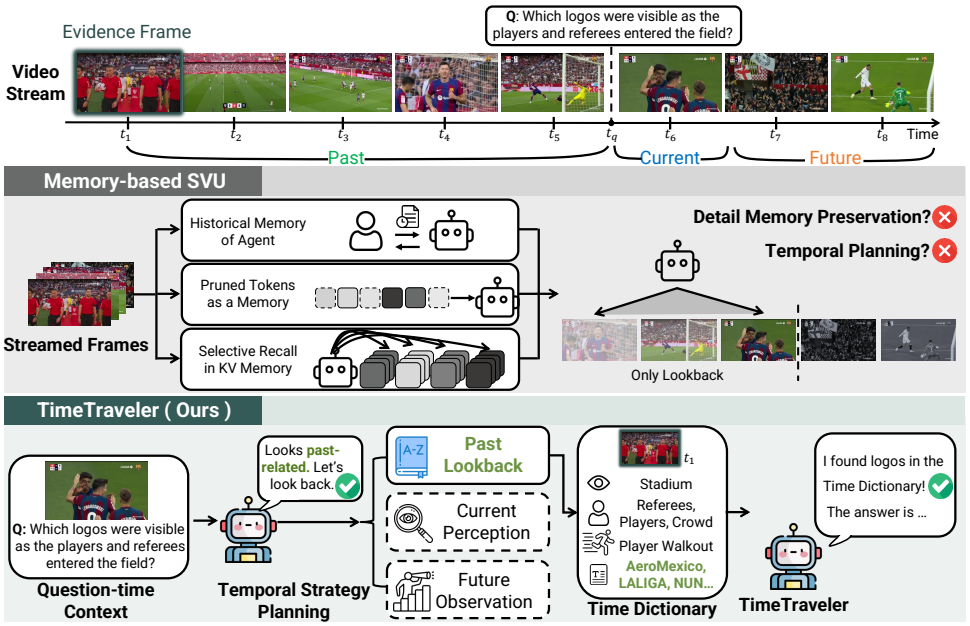


Figure 1: **Conceptual comparison between prior memory-based SVU and TIMETRAVELER.** Prior methods preserve streaming history as agentic memory [19], pruned tokens [22], or KV-memory recall [24], but rely on a single evidence-access path and do not ensure that retained evidence remains query-recoverable. In contrast, TIMETRAVELER plans whether each query should be grounded in the past, current scene, or future, and accesses a structured *Time Dictionary* accordingly.

information via token pruning [22, 25]. To make unfolding streams tractable, these methods often encode streamed observations into latent tokens, visual prototypes, or compressed representations for stable and efficient retention. Although these approaches are effective for streaming memory, such representations can obscure fine-grained cues including small text, object attributes, spatial relations, or transient actions, making them difficult to recover at question time, as shown in Fig. 1. Since future queries are unknown during observation, cues that seem incidental when stored may later become answer-critical. Thus, beyond retaining a compressed history, SVU requires an explicit, fine-grained, and queryable memory representation that keeps such details recoverable at question time.

Moreover, preserving recoverable evidence alone does not determine how that evidence should be accessed for an arbitrary user query. Prior approaches [19, 18, 25] largely overlook when the evidence for answering a query should be sought, without explicitly modeling the temporal structure of user queries. We argue that SVU is fundamentally a temporal grounding problem, where each query implicitly specifies a temporal scope $\{\text{past}, \text{current}, \text{future}\}$. Accordingly, current queries rely on local evidence, past queries require retrieval from accumulated observations, and future queries must be monitored until the relevant event occurs.

To bridge this gap, in this paper, we propose TIMETRAVELER, a SVU framework that stores incoming observations as an explicit, fine-grained, and queryable memory representation, and accesses the stored evidence according to the temporal scope for the given user query. Specifically, TIMETRAVELER is built on two components: (i) Time Dictionary, which stores the stream as lightweight structured per-frame captions that decompose visual information within each frame into multiple semantic fields. Each timestamp is associated with explicit textual evidence covering scenes, objects, actions, text, spatial relations, and contextual details, making the memory both directly queryable and efficient to maintain in real-time streaming settings, and (ii) Temporal Strategy Planner for deciding whether a query should be answered from past memory, the current scene, or future observation. The model then follows the selected temporal strategy: for past queries, it performs memory lookback; for current queries, it refers to structured perception of the current scene; and for future queries, it carries out structured observation until the required evidence becomes available, all within a unified SVU system.

Through experiments on two streaming benchmarks [26, 27], we show that TIMETRAVELER consistently outperforms various baselines. More importantly, the results support our central claim that SVU depends not only on retaining fine-grained history (Table 3), but also on accessing the right evidence according to the temporal intent of each query (Table 4). By combining temporal strategy planning with a structured Time Dictionary, TIMETRAVELER improves evidence access across past, current, and future contexts, enabling more effective question answering in video streams.

Our contributions are summarized as follows:

- We propose TIMETRAVELER, a streaming video QA framework that stores fine-grained evidence in a lightweight Time Dictionary and retrieves it through query-aware Temporal Strategy Planning. Unlike compressed visual memory, our design preserves query-recoverable evidence over long streaming horizons while supporting efficient updates.
- We formulate SVU as a temporal grounding problem and identify temporal routing as a missing inductive bias. We introduce Temporal Strategy Planning, a query-aware mechanism that selects how evidence should be acquired from past, current, or future contexts.
- We validate TIMETRAVELER across multiple backbones, streaming video benchmarks, and efficiency metrics, and provide mechanistic analyses showing that temporal strategy planning and the Time Dictionary improve evidence selection and streaming question answering quality.

2 Related Work

2.1 Large Vision-Language Models

Early LVLMs [28, 29, 30] show that visual instruction tuning can connect pretrained language backbones [31, 4] with visual encoders for general-purpose multimodal conversation. Subsequent models improve this paradigm by scaling model and data, refining visual tokenization, and strengthening vision-language alignment [2, 32, 5, 1]. Video-oriented LVLMs [33, 34, 35, 6, 3] further extend these capabilities to temporal inputs by treating videos as sequences of images and introducing video-specific connectors, temporal aggregation modules, or video instruction-tuning pipelines. Despite strong progress on video QA and captioning, most Video-LLMs are designed for offline or fixed-context inference, where the video is fully available before answering or has already been compressed into a finite context window. This assumption is restrictive for SVU, where frames arrive causally and a query may require evidence from past history, the current scene, or future observations.

2.2 Streaming Video Understanding

Streaming video understanding studies how multimodal models process continuously arriving visual evidence before the full video is available. A major line of work addresses the retention problem: how to maintain useful history from an unbounded stream under bounded computation and memory. StreamChat builds hierarchical memory for long-stream and multi-round interaction [36], Flash-VStream maintains compact memories for real-time long-video response [17], and StreamMem reduces streaming memory cost through visual cache compression [18]. Related memory-based or prototype-based systems summarize incoming observations into compact token memories, learned prototypes, or recurrent retained states [37, 19]. These methods make long-horizon processing practical by compressing accumulated context or reducing redundant visual tokens. In contrast, our framework keeps the retained history as explicit structured caption records: each sampled frame is decomposed into semantic fields such as actions, people, objects, scene context, text, and spatial relations, so the memory remains more structured and query-recoverable at question time rather than being collapsed into a compact visual state.

Another line of work focuses on online interaction and proactive response, asking when a streaming model should produce an output. VideoLLM-Online and related systems adapt Video-LLMs to interleaved video-text streams [21], while Dispider decomposes real-time interaction into perception, decision, and reaction modules [23]. StreamBridge converts offline Video-LLMs into proactive streaming assistants with a memory buffer and lightweight activation module [20], and QueryStream introduces query-aware pruning with relevance-triggered active response scheduling [22]. These approaches improve response timing and streaming efficiency, but they primarily decide whether or when the answer model should be activated, rather than explicitly modeling the temporal grounding

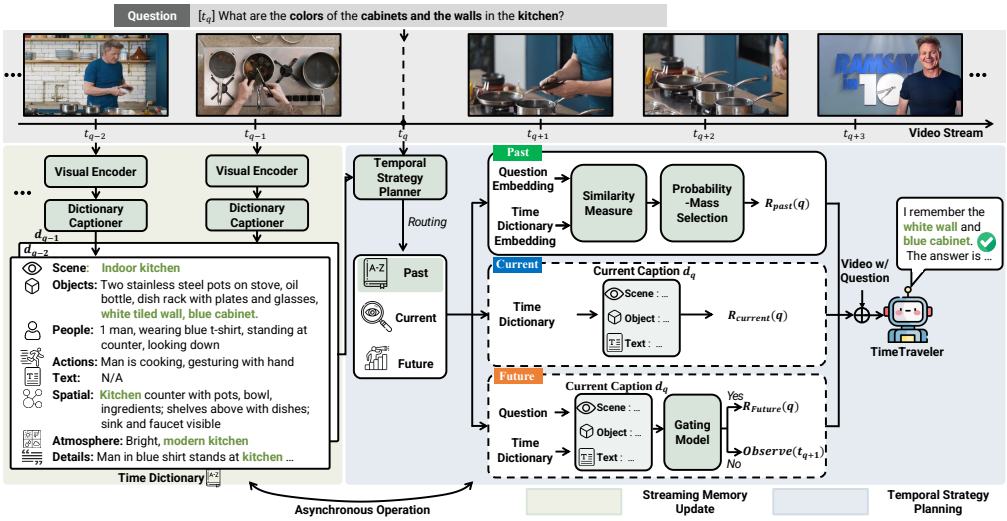


Figure 2: **Overview of TIMETRAVELER.** Incoming video frames are encoded into structured captions and stored in a timestamp-indexed Time Dictionary. Given a query, a Temporal Strategy Planner acquires evidence following temporal intent: past queries retrieve relevant captions through similarity-based lookback, current queries access local captions, and future queries continue observation until evidence appears. Then, the retrieved evidence is used to produce the final answer.

decision implied by the user's query. In contrast, our work makes this intermediate decision explicit: before assembling evidence, the system first determines. whether the query should be grounded in past memory, the current local context, or future observations, and then applies the corresponding evidence strategy over structured memory.

3 Proposed Method

Overview. We propose TIMETRAVELER, a streaming framework for query-conditioned evidence acquisition, as shown in Fig. 2. Our framework separates what to retain from the stream and how to access it for each query. Incoming frames are converted into structured caption records and stored in a causal Time Dictionary, together with sentence embeddings used only as lightweight retrieval handles. When a query arrives, a Temporal Strategy Planner selects one of three evidence: past, current, or future. These respectively retrieve prior evidence with recency-aware lookback, read the query-time caption, or continue observing until a response condition or horizon is reached.

3.1 Time Dictionary for Structured Evidence Retention

The goal of the Time Dictionary is to retain queryable evidence from a causal video stream without collapsing the history only into a visual prototype or recurrent hidden state. Rather than compressing observations into latent visual states, we represent the stream as a temporally indexed structured representation that preserves fine-grained, query-recoverable evidence. Let f_i denote the i-th sampled frame and τ_i its timestamp. At each observed moment, sampled at 1 fps in our streaming pipeline, a frozen captioner Φ_{cap} processes the frame once following eight fields: scene, objects, people, actions, text, spatial relations, atmosphere, and details. This produces a bounded-length structured caption $c_i = \Phi_{\text{cap}}(f_i)$, written in a consistent schema of the form:

$$c_i = \{\texttt{Scene} : s_i^{\text{scene}},\ \texttt{Objects} : s_i^{\text{obj}},\ \texttt{People} : s_i^{\text{people}},\ \texttt{Actions} : s_i^{\text{act}},\ \texttt{Text} : s_i^{\text{text}},\ \texttt{Spatial} : s_i^{\text{spatial}},\ \texttt{Atmosphere} : s_i^{\text{atm}},\ \texttt{Details} : s_i^{\text{Details}}\}. \tag{1}$$

Here, `Scene` summarizes locational and environmental context such as indoor or outdoor setting; `Objects` lists salient entities together with attributes such as color, size, and state; `People` describes count, appearance, clothing, and expression; `Actions` captures ongoing events and interactions among people and objects; `Text` records any visible signs, labels, subtitles, or logos; `Spatial` specifies layout and relative positions of key elements; `Atmosphere` summarizes lighting, color tone, time of day, weather, or mood; and `Details` provides a short coherent paragraph that captures

contextual cues not fully covered by the other fields. The schema encourages the captioner to explicitly expose fine-grained cues that may later become relevant to an arbitrary query.

For efficient lookback, the system also computes an auxiliary sentence embedding for each caption. A frozen text encoder Φ_{emb} encodes the full caption as one textual sequence and L2-normalizes it to e_i. For each sampled frame i, we create a dictionary entry $d_i = (i, \tau_i, c_i, e_i)$, and the Time Dictionary is then updated causally by appending the new entry, $\mathcal{D}_i = \mathcal{D}_{i-1} \| d_i$, with $\mathcal{D}_0 = ()$.

Thus, when a query arrives at frame index i_q, the system can only access $\mathcal{D}_{i_q}$, ensuring that no future frame is used for past or current answering. The Time Dictionary preserves streamed observations as explicit timestamped text records, allowing fine-grained evidence to remain available over long time horizons without being diluted as the stream unfolds. The auxiliary embeddings serve only as lightweight handles for efficient lookback, not as the primary memory content. For memory-efficient retrieval, we maintain these embeddings in a finite bin queue covering roughly the most recent 600 seconds of the stream, while the text-based Time Dictionary remains the persistent evidence store. This timestamp-indexed structure makes the accumulated stream addressable by different temporal strategies rather than forcing all queries to share a single compact memory state.

3.2 Query-Conditioned Temporal Strategy Planning

We then formulate temporal strategy selection as a query-conditioned policy over a discrete temporal action space $\{\text{past}, \text{current}, \text{future}\}$. Rather than predicting an answer directly, the policy determines how evidence should be acquired from the streaming setup. For example, a question such as "*What did the person pick up earlier?*" requires retrieving evidence from past memory, whereas "*What is the person doing now?*" is answered from the current scene. A future-facing instruction such as "*Tell me when the car stops.*" should not be answered immediately, but should instead be monitored until the requested condition becomes true or the observation horizon is reached. Temporal Strategy Planning captures these distinctions by assigning each query to the strategy that matches its temporal scope.

Temporal strategy planner. Given a query q issued at timestamp τ_q and latest observed frame index i_q, the planner serializes q together with a short recent-caption context and prompts a frozen lightweight instruction model to output exactly one temporal strategy $\{\text{past}, \text{current}, \text{future}\}$. The prompt instructs the model to choose `future` when the query asks the system to wait for a condition that has not yet occurred, `past` when the answer requires information outside the recent context, and `current` otherwise. The role of the planner is not to answer the question, but to decide how evidence should be sought before answer generation.

Past strategy: online probabilistic lookback. For past-directed queries, the system performs online probabilistic lookback over the observed Time Dictionary $\mathcal{D}_{i_q}$. Rather than selecting a fixed top-K set, the past strategy first forms a similarity-based candidate pool and then adaptively selects evidence using a recency-aware softmax distribution.

We first restrict the search to causal entries whose timestamps do not exceed the query time:

$$\mathcal{I}_q = \{i \mid d_i \in \mathcal{D}_{i_q},\ \tau_i \le \tau_q\}. \tag{2}$$

From this causal set, we compute a semantic similarity score $u_i = \cos(\phi(q), e_i)$ for each $i \in \mathcal{I}_q$, where $\phi(q)$ denotes the normalized query embedding. We keep the top-N entries by u_i as a candidate index set $\mathcal{C}_N(q) \subseteq \mathcal{I}_q$. In our implementation, we use $N = 32$.

For each candidate entry, we form a recency-aware selection logit:

$$\ell_i = u_i - \frac{\tau_q - \tau_i}{\tau_r}, \qquad i \in \mathcal{C}_N(q), \tag{3}$$

where τ_r controls how strongly older evidence is penalized. The logits are converted into a probability distribution with temperature T_{past}:

$$p_i = \frac{\exp(\ell_i / T_{\text{past}})}{\sum_{j \in \mathcal{C}_N(q)} \exp(\ell_j / T_{\text{past}})}, \qquad i \in \mathcal{C}_N(q). \tag{4}$$

We then sort candidate indices in descending order of p_i, denoted by $(1), (2), \ldots, (|\mathcal{C}_N(q)|)$, and select the smallest prefix whose cumulative probability mass exceeds a threshold t_p:

$$K^\star = \min \left\{ k : \sum_{m=1}^{k} p_{(m)} \geq t_p \right\}. \tag{5}$$

We define the past evidence set as $\mathcal{R}_{\text{past}}(q) = \{d_{(m)} \mid 1 \leq m \leq K^\star\}$. This cumulative-mass cutoff makes the number of selected past entries adaptive: concentrated distributions yield a small evidence set, while diffuse distributions retain more candidates. Importantly, embeddings are used only to construct the candidate pool and selection distribution; the evidence passed to the answer model is the stored caption text. This strategy supports non-local lookback when the answer depends on earlier observations that may no longer be visible.

Current strategy: query-time caption. For current-directed queries, global memory lookback is unnecessary and can introduce distractors from semantically similar but temporally irrelevant past events. The current strategy therefore reads the caption record at the query time as $\mathcal{R}_{\text{current}}(q) = \{d_{i_q}\}$. This provides direct access to the dictionary caption record at the query time and avoids mixing current evidence with temporally irrelevant past entries.

Future strategy: observe-or-respond gating. For future-directed queries, the system enters a deferred observation mode. Rather than invoking the answer model immediately, it periodically checks whether later evidence satisfies the query condition. Let $W \in \mathbb{N}_{>0}$ denote the evidence window size, i.e., the number of most recent dictionary entries passed to the gate. A frozen lightweight gate model g, prompted with a fixed yes/no instruction, takes the query and the window $\mathcal{D}_{i-W+1:i}$ and predicts whether the condition has been satisfied:

$$b_i = g(q, \mathcal{D}_{i-W+1:i}) \in \{0, 1\}, \qquad i > i_q. \tag{6}$$

The system polls the gate at intervals of $\Delta_{\text{poll}} \in \mathbb{N}_{>0}$ frames after i_q, and continues polling up to a maximum observation horizon $T_{\max}$ (in seconds) measured from the query time τ_q. The first polling index whose gate output is positive is selected:

$$i^\star = \min\{i > i_q : i \equiv i_q \pmod{\Delta_{\text{poll}}},\ \tau_i - \tau_q \leq T_{\max},\ b_i = 1\}. \tag{7}$$

If no polling index satisfies the condition before the maximum horizon $T_{\max}$, the system falls back to the latest available evidence under the observation budget. If no future evidence is available, it falls back to the caption record at the query time. This strategy enables delayed response without repeatedly calling the full answer model: the gate is evaluated during observation, and the answer model is invoked only once after the gate fires or the horizon is reached.

Evidence assembly. Let $\mathcal{R}_{\text{past}}(q)$, $\mathcal{R}_{\text{current}}(q)$, and $\mathcal{R}_{\text{future}}(q)$ denote the evidence returned by the three strategies. The final evidence sequence is selected according to the planned temporal strategy:

$$\mathcal{R}(q) = \begin{cases} \mathcal{R}_{\text{past}}(q), & \pi(q) = \text{past}, \\ \mathcal{R}_{\text{current}}(q), & \pi(q) = \text{current}, \\ \mathcal{R}_{\text{future}}(q), & \pi(q) = \text{future}. \end{cases} \tag{8}$$

The selected entries are serialized with their timestamps and captions, and are then passed to a frozen long-context answer model LM_{QA}, which generates the final response.

3.3 Streaming Inference

At inference time, the system runs an observe-and-respond loop. The ingest process continuously applies Φ_{cap} and Φ_{emb} to incoming frames and appends the resulting entries to $\mathcal{D}$. Streaming updates are efficient because each sampled frame produces one bounded caption record and one embedding append, while future queries require only lightweight periodic gating before a single final answer-generation step. When a query arrives, the dispatch process prompts the Temporal Strategy Planner once. Past and current queries invoke LM_{QA} immediately using evidence available at i_q. Future queries create a polling state and evaluate the prompt-based gate only at newly reached polling indices while the ingest process continues updating the Time Dictionary.

Table 1: Evaluation results on OVO-Bench. Overall accuracy is calculated as the macro average across taxonomies. Best results among open-source models are **boldfaced** and runner-up results are underlined. Proprietary model results are shown with gray background for reference.

Method	#Frames	Real-Time Visual Perception							Backward Tracing				Forward Act. Responding				Overall
		OCR	ACR	ATR	STU	FPD	OJR	Avg.	EPM	ASI	HLD	Avg.	REC	SSR	CRR	Avg.	Avg.
Human																	
Human	-	93.96	92.57	94.83	92.70	91.09	94.02	93.20	92.59	93.02	91.37	92.33	95.48	89.67	93.56	92.90	92.81
Proprietary Models (Offline)																	
Gemini 1.5 Pro [38]	1 fps	85.91	66.97	79.31	58.43	63.37	61.96	69.32	58.59	76.35	52.64	62.54	35.53	74.24	61.67	57.15	63.00
GPT-4o [39]	64	69.80	64.22	71.55	51.12	70.30	59.78	64.46	57.91	75.68	48.66	60.75	27.58	73.21	59.40	53.40	59.54
Open-Source Offline Models																	
LLaVA-Video-7B [33]	64	69.13	58.72	68.83	49.44	74.26	59.78	63.52	56.23	57.43	7.53	40.40	34.10	**69.95**	60.42	**54.82**	52.91
LLaVA-OV-7B [2]	64	66.44	57.80	73.28	53.37	71.29	61.96	64.02	54.21	55.41	21.51	43.71	25.64	67.09	58.75	50.50	52.74
LLaVA-N-Video-7B [40]	64	69.80	59.60	66.40	50.60	72.30	61.40	63.30	51.20	64.20	9.70	41.70	34.10	67.60	**60.80**	54.20	53.10
Qwen2-VL-7B [41]	64	60.40	50.46	56.03	47.19	66.34	55.43	55.98	47.81	35.48	**56.08**	46.46	31.66	65.82	48.75	48.74	50.39
InternVL-V2-8B [42]	64	67.11	60.55	63.79	46.07	68.32	56.52	60.39	48.15	57.43	24.73	43.44	26.50	59.14	54.14	46.60	50.15
LongVU-7B [43]	1 fps	55.70	49.50	59.50	48.30	68.30	63.00	57.40	43.10	66.20	9.10	39.50	16.60	69.00	60.00	48.50	48.50
Open-Source Online Models																	
Flash-VStream-7B [17]	1 fps	24.16	29.36	28.45	33.71	25.74	28.80	28.37	39.06	37.16	5.91	27.38	8.02	67.25	60.00	45.09	33.61
VideoLLM-Online-8B [21]	2 fps	8.05	23.85	12.07	14.04	45.54	21.20	20.79	22.22	18.80	12.18	17.73	-	-	-	-	-
VideoLLM-EyeWO [44]	1 fps	24.16	27.52	31.89	32.58	44.55	35.87	32.76	39.06	38.51	6.45	28.00	-	-	-	-	-
Dispider-7B [23]	1 fps	57.72	49.54	62.07	44.94	61.39	51.63	54.55	48.48	55.41	4.30	36.06	18.05	37.36	48.75	34.72	41.78
TimeChat-Online-7B [25]	1 fps	69.80	48.60	64.70	44.90	68.30	55.40	58.60	53.90	62.80	9.10	42.00	32.50	36.50	40.00	36.40	45.60
StreamAgent-7B [24]	1 fps	71.20	53.20	63.60	53.90	67.30	58.70	61.30	54.80	58.10	25.80	41.70	**35.90**	48.40	52.00	45.40	49.40
QueryStream-7B [22]	1 fps	74.50	47.70	70.70	46.60	71.30	57.60	61.40	54.20	63.50	8.60	42.10	33.20	43.10	40.80	39.03	47.51
InternVL3.5-8B [5]	1fps	79.87	65.14	69.83	**61.80**	70.30	65.22	68.69	33.00	59.46	4.30	32.25	32.66	60.10	35.00	42.59	47.84
+ TIMETRAVELER	1fps	**91.95**	70.64	75.00	**61.80**	**75.25**	**77.17**	**75.30**	33.67	60.81	9.68	34.72	22.21	50.24	38.75	37.06	49.03
Qwen3-VL-8B [16]	1fps	75.17	60.55	66.38	53.93	66.34	57.07	63.24	52.53	67.57	10.22	43.44	33.09	65.50	46.67	48.42	51.70
+ TIMETRAVELER	1fps	87.25	**73.39**	**76.72**	59.55	68.32	74.46	73.00	**59.26**	**69.59**	25.27	**51.66**	32.23	67.25	48.75	49.41	**58.02**

Once the gate fires, or the maximum horizon is reached, the system invokes $\mathrm{LM}_{\mathrm{QA}}$ exactly once with the selected future evidence. At QA time, the answer model takes as input the query q, the video observations available at decision time, and the temporal evidence assembled by our method, $\mathcal{R}(q)$, and produces the final VQA answer, $a = \mathrm{LM}_{\mathrm{QA}}\big(v, q, \mathcal{R}(q)\big)$, where v denotes the video frames available when the system answers the query and a is the generated answer.

4 Experiments

4.1 Experimental Setting

Benchmarks and Metrics. We evaluate TIMETRAVELER on two complementary streaming video understanding benchmarks: OVO-Bench [27] and StreamingBench [26]. OVO-Bench measures three challenging capabilities in online video understanding: real-time visual perception, backward tracing, and forward active responding. Following the benchmark protocol, we report category-wise accuracies together with the macro average across the three taxonomies as the overall score. StreamingBench evaluates streaming comprehension over 18 tasks spanning real-time visual understanding, omni-source understanding, and contextual understanding.

Implementation Details. TIMETRAVELER builds on the Qwen3 family [16]: it uses a frozen 8-bit-quantized Qwen3-VL-2B as both the visual captioner and the planner/gate model, Qwen3-Embedding-0.6B as the text encoder, and Qwen3-VL-8B as the frozen answer model; we also evaluate with InternVL3.5-8B [5]. Incoming frames are sampled at 1 FPS and converted into structured caption records with eight semantic fields, while one auxiliary embedding per caption is maintained only as a retrieval handle in a finite cache covering roughly the most recent 600 seconds. For past-directed queries, we use a candidate pool size of $N = 32$ and set $\tau_r = 150$, $T_{\mathrm{past}} = 0.02$, and $t_p = 0.70$; for future-directed queries, we set $\Delta_{\mathrm{poll}} = 1$ frame, $T_{\max} = 30$ s, and $W = 1$. When InternVL3.5 serves as the answer model, we supply maximum 128 frames as context limits; all other settings remain identical. All components are used in frozen form and only prompt-based planning is applied at inference time. Experiments are run on a single NVIDIA RTX PRO 6000 Blackwell GPU.

4.2 Comparison with State-of-the-art Methods

Tables 1 and 2 present results on OVO-Bench and StreamingBench, respectively. TIMETRAVELER achieves the best overall performance among open-source models on both benchmarks, consistently improving both Qwen3-VL (51.70→58.02 on OVO-Bench; 59.33→64.51 on StreamingBench) and InternVL3.5 (47.84→49.03; 61.71→65.64). The gains align with our two design goals: the Time Dictionary improves detail-sensitive tasks (e.g., OCR, TR) where fine-grained cues must be preserved, while temporal strategy planning benefits temporally grounded tasks (e.g., EPM, PO) that require selecting evidence from the right temporal scope.

Table 2: Performance comparison on StreamingBench. Best results among open-source models are **boldfaced** and runner-up results are underlined.

Model	#Frames	Real-Time Visual Understanding											Omni-Source Understanding					Contextual Understanding					Overall
		OP	CR	CS	ATP	EU	TR	PR	SU	ACP	CT	All	ER	SCU	SD	MA	All	ACU	MCU	SQA	PO	All	
Human																							
Human‡	-	89.47	92.00	93.60	91.47	95.65	92.52	88.00	88.75	89.74	91.30	91.46	88.00	88.24	93.60	90.27	90.26	88.80	90.40	95.00	100.00	93.55	91.66
Proprietary MLLMs																							
Gemini 1.5 pro [38]	1 fps	79.02	80.47	83.54	79.67	80.00	84.74	77.78	64.23	71.95	48.70	75.69	46.80	39.60	74.90	80.00	60.22	51.41	40.73	54.80	45.10	48.73	67.07
GPT-4o [39]	64	77.11	80.47	83.91	76.47	70.19	83.80	66.67	62.19	69.12	49.22	73.28	41.20	37.20	43.60	56.00	44.50	41.20	38.40	32.80	56.86	38.70	60.15
Claude 3.5 Sonnet [45]	20	80.49	77.34	82.02	81.73	72.33	75.39	61.11	61.79	69.32	43.09	72.44	31.60	34.00	32.80	48.80	36.80	38.40	34.80	34.40	64.71	37.70	57.68
Open-Source Offline Video MLLMs																							
LLaVA-OV-7B [2]	32	80.38	74.22	76.03	80.72	72.67	71.65	67.59	65.45	65.72	45.08	71.12	40.80	**37.20**	33.60	44.80	38.40	35.60	36.00	27.27	29.55	32.74	56.36
Qwen2-VL-7B [41]	0.2–1 fps†	75.20	82.81	73.19	77.45	68.32	71.03	72.22	61.19	61.47	46.11	69.04	41.20	22.00	32.80	43.60	34.90	31.20	26.00	39.60	22.73	31.66	54.14
MiniCPM-V 2.6-8B [46]	32	71.93	71.09	77.92	75.82	64.60	65.73	70.37	56.10	62.32	53.37	67.44	40.80	24.00	34.00	41.20	35.00	34.00	31.60	41.92	22.22	34.97	53.85
InternVL-V2-8B [42]	16	68.12	60.94	69.40	77.12	67.70	62.93	59.26	53.25	54.96	56.48	63.72	37.60	26.40	37.20	42.00	35.80	32.00	31.20	32.32	40.91	32.42	51.40
Kangaroo-7B [47]	64	71.12	**84.38**	70.66	73.20	67.08	61.68	56.48	55.69	62.04	38.86	64.60	37.60	31.20	28.80	39.20	34.20	32.80	26.40	33.84	16.00	30.06	51.10
LongVA-7B [48]	128	70.03	63.28	61.20	70.92	62.73	59.50	61.11	53.66	54.67	34.72	59.96	39.60	32.40	28.00	41.60	35.40	32.80	29.60	30.30	15.91	29.95	48.66
VILA-1.5-8B [49]	14	53.68	49.22	70.98	56.86	53.42	53.89	54.63	48.78	50.14	17.62	52.32	41.60	26.40	28.40	36.00	33.10	26.80	34.00	23.23	17.65	27.35	43.20
Video-CCAM-14B [50]	96	56.40	57.81	65.30	62.75	64.60	51.40	42.59	47.97	49.58	31.61	53.96	33.60	22.00	28.40	34.80	29.70	27.60	24.40	16.67	22.73	22.88	42.53
Video-LLaMA2-7B [51]	32	55.86	55.47	57.41	58.17	52.80	43.61	39.81	42.68	45.61	35.23	49.52	30.40	32.40	30.40	36.00	32.40	24.80	26.80	18.67	0.00	21.93	40.40
Open-Source Online Video MLLMs																							
Flash-VStream-7B [17]	-	25.89	43.57	24.91	23.87	27.33	13.08	18.52	25.20	23.87	48.70	23.23	25.91	24.90	25.60	28.40	26.00	24.80	25.20	26.80	1.96	24.12	24.04
VideoLLM-online-8B [21]	2 fps	39.07	40.06	34.49	31.05	45.96	32.40	31.48	34.16	42.49	27.89	35.99	31.20	26.51	24.10	32.00	28.45	24.19	29.20	30.80	3.92	26.55	32.48
Dispider-7B [23]	1 fps	74.92	75.53	74.10	73.08	74.44	59.92	76.14	62.91	62.16	45.80	67.63	35.46	25.26	38.57	43.34	35.66	39.62	27.65	34.80	25.34	33.61	53.12
TimeChat-Online-7B [25]	1 fps	80.22	82.03	79.50	83.33	76.10	78.50	78.70	64.63	69.60	57.98	75.36	–	–	–	–	–	–	–	–	–	–	–
QueryStream-7B [22]	1 fps	**82.38**	**84.38**	79.18	82.37	77.99	81.31	78.70	65.04	69.32	47.34	75.32	–	–	–	–	–	–	–	–	–	–	–
InternVL3.5-8B [5]	1fps	78.86	82.81	85.17	81.41	77.99	79.75	83.33	70.73	76.99	55.32	77.60	40.40	30.80	42.80	53.60	41.90	42.40	36.40	**48.40**	40.00	41.80	61.71
+ TIMETRAVELER	1fps	82.11	82.81	**89.27**	84.62	**83.65**	87.54	87.04	**73.17**	**77.56**	53.72	**80.72**	**48.40**	34.40	**48.00**	51.20	**45.50**	**47.60**	**44.80**	46.80	53.23	48.10	**65.64**
Qwen3-VL-8B [16]	1fps	75.10	76.60	80.10	81.10	77.40	71.70	77.80	61.40	69.30	55.90	72.80	41.20	32.00	46.80	**55.20**	43.80	40.00	31.60	34.40	52.40	36.75	59.33
+ TIMETRAVELER	1fps	81.84	79.69	87.70	**85.58**	76.73	**90.97**	**87.96**	69.11	75.28	**59.04**	80.16	43.20	26.40	44.00	49.20	40.70	44.40	41.60	48.00	**62.90**	**49.20**	64.51

Table 3: **Comparison of memory representations under an equal token budget on OVO-Bench.** #Tokens denotes the average number of memory tokens injected into the answer model at inference time. Qwen3-VL-8B is used as the backbone.

Memory Type	**#Tokens**	**Real-Time Acc.**	**Backward Acc.**	**Forward Acc.**	**Overall Acc.**
No Memory	–	63.24	43.44	48.42	51.70
Visual Prototype Memory	203.11	66.19	44.16	**51.36**	53.90
Flat Text Memory	236.60	70.98	48.57	47.83	55.79
Structured Dictionary (Ours)	245.23	**73.00**	**51.66**	49.41	**58.02**

4.3 Effect of TIMETRAVELER

Effect of Structured Time Dictionary. Table 3 compares alternative memory representations under a similar memory-token budget on queries that require recovering fine-grained past visual details. We compare three memory designs: *visual prototype memory*, which clusters accumulated visual tokens into K representative prototypes via k-means over the streaming sequence (K=2, configured to match the token budget of our method for fair comparison); *flat text memory*, which stores a single unstructured free-form caption per frame; and our *structured dictionary*, which decomposes each caption into eight semantic fields. Relative to the no-memory baseline, both visual prototype memory and flat text memory improve performance, confirming that retaining historical information is essential for streaming video QA. However, the proposed structured dictionary achieves the strongest overall result, improving the direct Qwen3-VL baseline from 51.70 to 58.02 and outperforming flat text memory by 2.23 points overall, with consistent gains in both real-time and backward accuracy. This suggests that the benefit does not come merely from storing more past content, but from organizing memory into semantically typed fields that preserve fine-grained details while making relevant evidence easier to retrieve.

Effect of Temporal Strategy Planning. Table 4 provides direct evidence for temporal strategy planning. While all memory-based settings outperform the no-memory baseline, our routing achieves the best backward-tracing average (51.66) with top scores on EPM (59.26) and ASI (69.59). Notably, indiscriminate lookback is not universally beneficial: some backward-tracing questions still depend on current grounding, and always retrieving past evidence can introduce temporally irrelevant distractors. Query-conditioned routing preserves the benefit of memory retrieval while avoiding unnecessary access to irrelevant history, confirming that streaming VQA requires selecting the appropriate temporal strategy per query, not just retaining past evidence.

4.4 Ablation Study on Time Dictionary Schema

Table 5 reports a schema-group ablation that removes one semantic field group at a time while keeping all other components unchanged. The full schema achieves the best overall performance on both benchmarks. Removing *Explicit Text* causes the largest drop on Text-Rich Understanding (TR: 90.97 $\rightarrow$ 85.98) and Optical Character Recognition (OCR: 87.25 $\rightarrow$ 83.22), confirming that fine-grained

Table 4: **Effect of temporal routing on OVO-Bench Backward Tracing.** EPM: Episode Memory, ASI: Action Sequence Identification, HLD: Highlight Detection.

Setting	Backward Tracing			
	EPM	HLD	ASI	Avg.
Baseline (no memory) [16]	52.53	10.22	67.57	43.44
All-to-Past Lookback	57.91	**25.81**	67.57	50.71
Random Routing	57.94	25.13	66.52	50.28
Our Routing	**59.26**	25.27	**69.59**	**51.66**

Table 5: **Schema-group ablation for the Time Dictionary.** Each row removes one of four schema groups while keeping all else fixed. Overall is computed only over the listed subtasks, not over the full benchmark.

Setting	OVO-Bench			StreamingBench		
	OCR	OJR	Overall	TR	SU	Overall
Baseline [16]	75.17	57.07	66.12	71.70	61.40	67.23
Full Schema (Ours)	87.25	74.46	80.86	90.97	69.11	81.48
w/o Background	83.89	69.57	76.73	90.97	69.11	81.48
w/o Entity-Centric	83.89	72.83	78.36	91.59	67.07	80.95
w/o Explicit Text	83.22	73.37	78.30	85.98	68.29	78.31
w/o Contextual Sent.	85.23	71.74	78.49	89.41	67.07	79.72

on-screen cues such as signs, logos, and labels require a dedicated field to remain recoverable at query time. Removing *Entity-Centric Dynamics* (Objects + People + Actions + Spatial) degrades OCR (87.25 $\rightarrow$ 83.89), Object Recognition (OJR: 74.46 $\rightarrow$ 69.57), and Spatial Understanding (SU: 69.11 $\rightarrow$ 67.07), showing that object-level attributes and inter-entity relations likewise benefit from explicit decomposition. The remaining two groups (Background, Contextual Sentence) also contribute to overall performance, but their removal patterns are less task-specific; we attribute this to the fact that the eight schema fields are not fully independent, and the model leverages overlapping cues across fields to partially compensate for a missing group. Altogether, these results support our design: structured decomposition into typed fields preserves fine-grained evidence that would otherwise be lost in a flat or compressed representation.

4.5 Asynchronous Streaming and Query Latency

Streaming-state updates and query answering are executed asynchronously and therefore should not be collapsed into a single end-to-end latency number. Under a 1 fps input stream, maintaining the Time Dictionary requires 637.51 ms per second of video, which stays within the real-time budget and leaves headroom for continuous online operation. Once this state has been prepared, query-time reasoning introduces only modest overhead over direct visual QA: the total latency is 578.25 ms for past queries, 557.62 ms for current queries, and 565.11 ms for future queries with one polling step, compared with 547.22 ms for the visual-only baseline, corresponding to only 10–31 ms of additional delay depending on the temporal branch. This breakdown reflects the practical advantage of our design: the heavier video processing cost is amortized during streaming, while user-time inference operates on compact structured state, allowing responsive query answering without reprocessing the full visual history for each question. A detailed latency breakdown is provided in Table 6.

5 Discussion and Conclusion

Limitation. TimeTraveler relies on caption-based textual evidence as the persistent memory representation. While this improves query recoverability and efficiency, it may inherit errors from the captioner, especially for small objects, ambiguous actions, or OCR under low resolution. The current planner and future gate are prompt-based frozen models, so misclassification of temporal scope can lead to suboptimal evidence selection. Finally, broader validation under noisier real-world streams, multi-turn interactions, and longer future waiting horizons remains future work.

Conclusion. We present TIMETRAVELER, a streaming video understanding framework built on two complementary ideas: storing past observations in an explicit and queryable structure, and matching each query to the appropriate temporal scope. TIMETRAVELER maintains a lightweight Time Dictionary that records each observed moment as structured textual evidence, preserving fine-grained cues in a more recoverable form. On top of this memory, a Temporal Strategy Planner determines whether the answer should be grounded in past observations, the current scene, or future evidence, enabling query-specific evidence access. Extensive experiments and analyses on OVO-Bench and StreamingBench show that this combination improves both detail-sensitive and temporally grounded reasoning, while maintaining practical efficiency under asynchronous streaming deployment.

Acknowledgments

This work was supported in part by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. RS-2024-00334321), by the MSIT (Ministry of Science and ICT), Korea, under the ITRC (Information Technology Research Center) support program (No. IITP-2026-RS-2023-00258649) supervised by the IITP (Institute for Information and Communications Technology Planning and Evaluation), and by Global - Learning & Academic research institution for Master's·PhD students, and Postdocs(G-LAMP) Program of the National Research Foundation of Korea(NRF) grant funded by the Ministry of Education(No. RS-2025-25442355), by Institute of Information and Communications Technology Planning and Evaluation (IITP) grant funded by the Korea government (MSIT) (No. RS-2022-II220078, RS-2025-02653113).

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Table 6: **Asynchronous latency breakdown of TIMETRAVELER.** The two blocks run independently: stream processing executes continuously, while query processing is triggered on demand. Stream latency is normalized to one second of input at 1 fps. Qwen3-VL-8B is used as the backbone.

Stream Processing (always-on)		
Stage	**Latency**	**TFLOPs**
Video decode	15.72 ms	–
Captioner	605.82 ms	1.53
Embedder	15.97 ms	0.20
Total (per 1 s)	**637.51 ms**	**1.73**

Query Processing (on-demand)			
Path	**Stage**	**Latency**	**TFLOPs**
Baseline	QA (Visual-only)	547.22 ms	1.42
Our Past	Scope planning	9.60 ms	1.39
	Retrieval	15.62 ms	0.03
	QA (8 cap)	553.03 ms	22.45
	Total	**578.25 ms**	**23.87**
Our Current	Scope planning	9.60 ms	1.39
	QA (1 cap)	548.02 ms	3.81
	Total	**557.62 ms**	**5.2**
Our Future	Scope planning	9.60 ms	1.39
	Gating ($\times$1; +7.49 ms/extra)	7.49 ms	5.73
	QA (1 cap)	548.02 ms	3.81
	Total	**565.11 ms**	**10.93**

Table 7: **Memory footprint comparison across memory designs on Qwen3-VL-8B [16].** Visual tokens per frame are measured at 240 for inputs resized to `max_pixels` $= 401408$. Visual prototype uses K=32 slots. Our structured dictionary stores 800-byte UTF-8 caption text and a 1024-dim FP16 embedding per frame. Retrieval embeddings are maintained in a finite queue covering the most recent 600 s, so their cost is capped beyond 10 minutes.

Memory Design	Per-frame	1 min	5 min	10 min	30 min	1 hour
Visual Full KV Cache	33.75 MB	1.98 GB	9.89 GB	19.78 GB	59.33 GB	118.65 GB
Visual Hidden State	1.88 MB	112.5 MB	562.5 MB	1.10 GB	3.30 GB	6.59 GB
Visual Prototype (K=32)	–	1.6 MB	1.6 MB	1.6 MB	1.6 MB	1.6 MB
Time Dictionary (Ours)						
Caption Text	0.78 KB	46 KB	234 KB	469 KB	1.4 MB	2.8 MB
Retrieval Embedding	2.00 KB	120 KB	600 KB	1.2 MB	1.2 MB	1.2 MB
Total	2.78 KB	167 KB	834 KB	1.6 MB	2.6 MB	4.0 MB

A Additional Analysis of TIMETRAVELER

A.1 Memory footprint

Table 7 quantifies the storage cost that underlies our design choice of textual over visual memory. A central motivation of TIMETRAVELER is that streaming memory should remain both *fine-grained* and *lightweight*; visual representations can preserve perceptual detail but scale linearly with stream length at a prohibitive rate, while compressed prototypes(K=32 slots, constructed via k-means clustering of visual tokens) stay compact but sacrifice queryable detail. Our structured dictionary occupies only 2.78 KB per frame—roughly $690\times$ less than a visual hidden state and over $12{,}000\times$ less than a full KV cache—yet retains semantically typed, directly queryable evidence. Over a one-hour stream at 1 FPS, the entire dictionary amounts to 4.0 MB, whereas visual KV memory would require over 118 GB. Even the most compact visual baseline, prototype memory with K=32 slots, achieves its small footprint only by collapsing the stream into a fixed-size state that discards temporal indexing and fine-grained cues. In contrast, our dictionary scales linearly but remains practically negligible, keeping every observed moment individually addressable while adding virtually no system-level memory burden.

A.2 Sensitivity to temperature T.

Table 8 examines the temperature T used in the recency-aware softmax for past-evidence selection. Since T is a hyperparameter of the past-lookback module, Table 8 reports overall backward accuracy only. Results are obtained by applying our method on top of Qwen3-VL. The best backward-tracing accuracy is obtained at $T = 0.02$, which reaches 51.66. When the temperature is reduced to 0.01,

Table 8. **Sensitivity to temperature T on OVO Backward Tracing.**

T	Backward Avg.
0.01	50.39
0.02	**51.66**
0.05	50.34
0.10	50.16

Table 9. **Sensitivity to threshold t_p on OVO Backward Tracing.**

t_p	Backward Avg.
0.10	50.72
0.30	50.50
0.50	50.84
0.70	**51.66**
0.90	50.74
1.00	50.07

Table 10. **Sensitivity to recency scale τ on OVO Backward Tracing.**

τ	Backward Avg.
No recency bias	49.51
150	**51.66**
300	51.35
600	51.51
1200	50.87

performance drops to 50.39, suggesting that overly sharp distributions can over-commit to a very small set of candidates and become brittle to retrieval noise. Increasing the temperature to 0.05 or 0.10 also hurts performance, indicating that excessively smooth selection weakens the model's ability to focus on the most relevant historical evidence. These results support our choice of a moderately sharp temperature that balances selectivity and robustness.

A.3 Sensitivity to nucleus threshold t_p.

Table 9 studies the nucleus threshold t_p, which controls how much cumulative probability mass is retained in the adaptive evidence prefix. Since t_p is a hyperparameter of the past-lookback module, Table 9 reports overall backward accuracy only. Results are obtained by applying our method on top of Qwen3-VL. The strongest result again appears at $t_p = 0.70$, yielding 51.66 backward accuracy. Smaller thresholds such as 0.10 or 0.30 underperform, which suggests that overly aggressive truncation discards useful complementary context. On the other hand, very large thresholds such as 0.90 or 1.00 also degrade performance, implying that including too many lower-confidence entries introduces distracting historical evidence. This trend confirms that adaptive selection is most effective when it retains a compact but sufficiently informative subset of past captions.

A.4 Sensitivity to recency scale τ.

Table 10 evaluates the recency scale τ, which determines how strongly older evidence is penalized during lookback. Since τ is a hyperparameter of the past-lookback module, Table 10 reports overall backward accuracy only. Results are obtained by applying our method on top of Qwen3-VL. Setting $\tau = 150$ gives the highest backward average of 51.66, while removing recency bias entirely (implemented by dropping the second term in Eq. (4)) leads to a clear drop to 49.51. This gap shows that semantic similarity alone is insufficient for streaming QA, because temporally distant but semantically related captions can act as distractors. Larger values such as 300 and 600 remain competitive but do not surpass the proposed setting, and performance declines again at 1200, where the temporal penalty becomes too weak. In particular, the competitive result at $\tau = 600$ suggests that retrieval quality remains strong even over roughly 10 minutes of context. Overall, the results validate recency-aware retrieval as an important ingredient for temporally grounded evidence acquisition.

A.5 Adaptive selection vs. fixed top-K.

Finally, Table 11 compares our online probabilistic lookback with a fixed top-K retrieval variant under the same routing policy. We set $K = 4$ so that the fixed-retrieval baseline uses a memory budget comparable to the average number of entries selected by our online probabilistic method. Replacing the adaptive cumulative-mass selection rule with this fixed top-K strategy reduces backward accuracy from 51.66 to 49.64. This result shows that the number of relevant past entries should not be treated as constant across queries: some questions are resolved by a small amount of concentrated evidence, whereas others require a broader set of supporting captions. Our probabilistic selection mechanism adapts to this variability, preserving compactness when confidence is concentrated while expanding the evidence set when the retrieval distribution is more diffuse.

A.6 Effect of gating on proactive output task.

Table 12 examines the effect of future gating on StreamingBench Proactive Output. In this task, a question is posed before the answer-relevant event, and the model is evaluated at one second after

Table 11: **Effect of adaptive past-evidence selection on OVO Backward Tracing.** Since this comparison isolates the past-lookback module, we report overall backward accuracy only. Results are obtained by applying our method on top of Qwen3-VL-8B[16]. We compare the proposed online probabilistic lookback with a fixed top-K variant under the same routing setup. Replacing adaptive selection with fixed top-K retrieval degrades performance.

Selection Strategy	Backward Avg.
Top-K Selection ($K = 4$)	49.64
Online Probabilistic Selection (Ours)	**51.66**

Table 12: **Effect of the future gating on StreamingBench Proactive Output.**

Backbone	Method	Acc.
Qwen3-VL-8B	Baseline	52.40
	+ Time Dictionary w/o gating	45.20
	+ Time Dictionary w/ gating (Ours)	**62.90**
InternVL3.5-8B	Baseline	40.00
	+ Time Dictionary w/o gating	41.20
	+ Time Dictionary w/ gating (Ours)	**53.23**

the ground-truth moment on whether it correctly decides to answer and, if so, whether the answer is correct. The baseline makes this decision using only the current visual frames, while our settings additionally provide the Time Dictionary memory. Without gating, the 8B answer model itself judges whether to respond, but this actually hurts Qwen3-VL (52.40→45.20), suggesting that the model struggles to make reliable respond-or-wait decisions when given the full memory context. With gating, a lightweight 2B model sees only the current-frame caption and the question to make the respond-or-wait decision, then hands off to the 8B model for answering. This yields substantial gains on both backbones (Qwen3-VL: +10.50; InternVL3.5: +13.23), confirming that separating the gating decision from the answering model is more effective.

A.7 Textual Input Sensitivity

To verify that the Time Dictionary provides substantive evidence rather than merely additional context, we conduct a noise injection experiment (Fig. 3). We replace a fraction ρ of the stored captions with irrelevant captions sampled from unrelated videos and evaluate on OVO-Bench. Performance degrades steadily as ρ increases, dropping from 58.02 at ρ=0 to 50.27 at ρ=1.0, which falls below the no-memory baseline (51.70). This shows that the answer model actively relies on the retrieved caption content to produce correct answers, rather than benefiting from the mere presence of additional text input. It also validates the quality of our Time Dictionary: the captions generated by the streaming captioner carry accurate, answer-relevant evidence that meaningfully contributes to question answering.

A.8 Qualitative Result of Lookback and Dictionary.

Fig. 4 illustrates a representative lookback retrieval case. The user asks which muscle groups are highlighted during a front raise exercise. The similarity score peaks at a past frame showing a 3D anatomy diagram with red-highlighted deltoids, and the corresponding Time Dictionary caption retains the precise on-screen text, object descriptions, and spatial relations needed to answer. This example demonstrates that our structured captions preserve the fine-grained evidence that compressed visual representations would likely discard, and that the retrieval mechanism successfully surfaces the most relevant past observation.

B Prompts in TIMETRAVELER

We document the three prompt templates used in our pipeline: the per-frame **captioner** that populates the Time Dictionary (§B.1), the **planner** that performs temporal routing (§B.2), and the **gating** model that decides cue-trigger emission for proactive-output questions (§B.3).

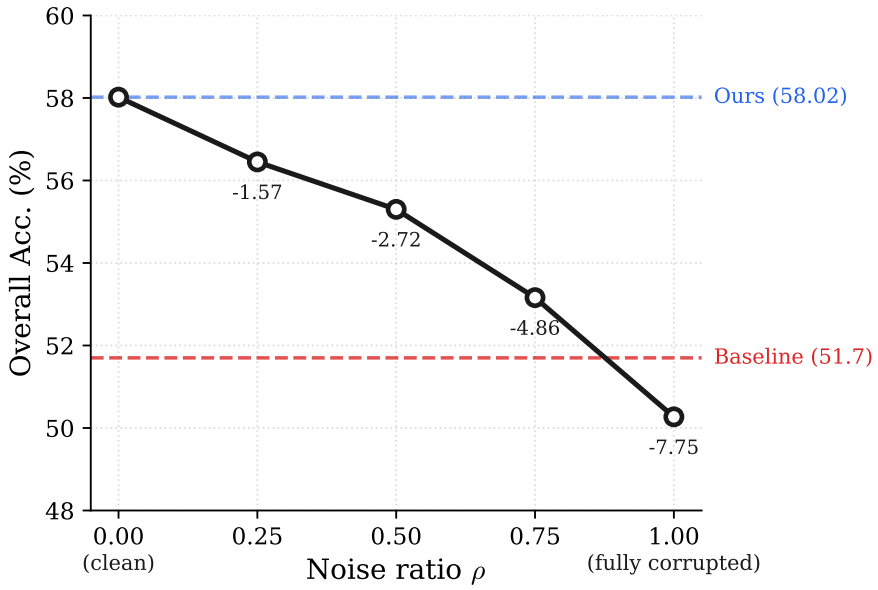


Figure 3: **Textual input sensitivity on OVO-Bench.** We replace a fraction ρ of the Time Dictionary captions with irrelevant captions sampled from unrelated videos and measure the resulting Overall accuracy. Dashed lines indicate the baseline (no memory) and our full method (clean captions).

B.1 Captioner Prompt

Per-frame captions populating the Time Dictionary are produced by a lightweight VLM (8-bit-quantized Qwen3-VL-2B [16]) using the structured 8-field template below. The prompt enforces concision, non-redundancy across fields, and a fixed field order so that downstream schema-level operations (e.g., field-routed retrieval, schema-drop ablation) can address each field deterministically. Fields with no observable content are filled with the literal string "N/A".

Captioner Prompt

```
Analyze the image and provide a structured description using the format below.
Be concise. List key items only, do not repeat information across fields. Focus
on the most salient elements. If a field is not applicable, write 'N/A'.

Scene: [locational or environmental information, indoor/outdoor]
Objects: [key objects with attributes such as color, size, state]
People: [count, appearance, clothing, expression]
Actions: [what is happening, interactions between people/objects]
Text: [any visible text, signs, labels, subtitles, logos]
Spatial: [layout and relative positions of key elements]
Atmosphere: [lighting, color tone, time of day, weather, mood]
Details: [description sentences that are shown in the image in one coherent
paragraph. Write 2-3 sentences and keep it concise, around 80-120 words.]
```

Decoding configuration. Greedy decoding with temperature $= 0$, top-p $= 0.8$, top-k $= -1$, presence penalty $= 1.5$, repetition penalty $= 1.0$, and max new tokens $= 400$.

B.2 Planner Prompt

Each incoming question is routed into one of three temporal regions (`past`, `current`, or `future`) by a caption-augmented meta-classifier (8-bit-quantized Qwen3-VL-2B, text-only inference). In addition to the question, the planner receives the *current frame caption* – the most-recent caption from the Time Dictionary – so that routing decisions are grounded in present visual context. The decision selects the retrieval scope used downstream: specific past captions retrieved by query similarity

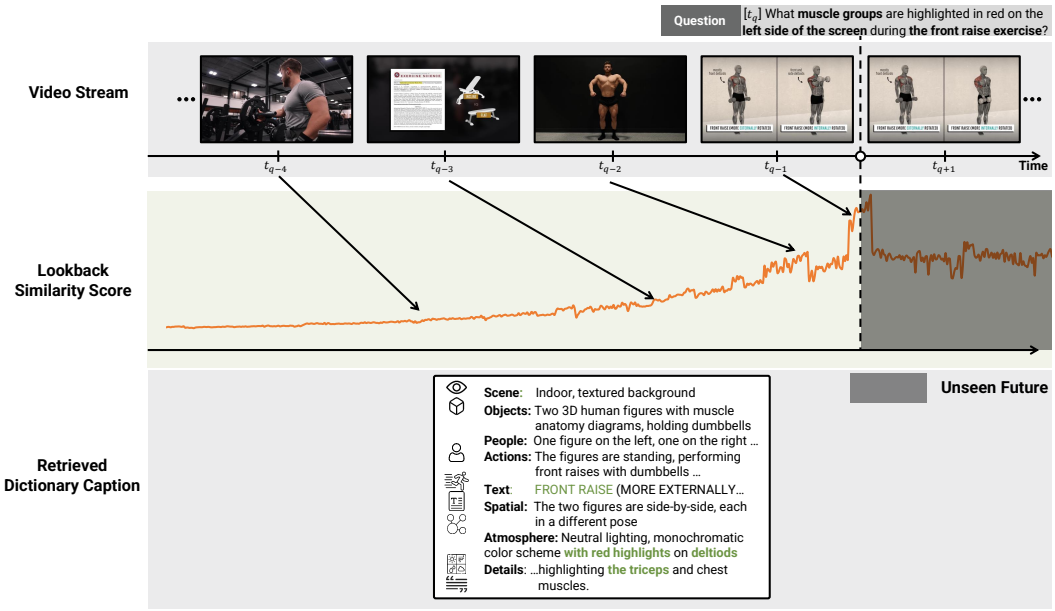


Figure 4: Qualitative example of lookback retrieval on OVO-Bench. Given a query at time t_q, the similarity score over past captions (orange curve) peaks at the frame most relevant to the question. The retrieved Time Dictionary caption preserves fine-grained details such as on-screen text (*"FRONT RAISE"*), spatial layout, and highlighted muscle groups, which are essential for answering correctly.

(`past`), the most-recent frame caption from the temporal buffer (`current`), or a wait-and-emit gating loop (`future`).

Planner Prompt (curcap v11)

```
You route streaming video questions into one of three temporal regions based on
what the answer depends on.
• current: the answer is a property of the frame at the moment of asking. It
does not require looking back or aggregating earlier content.
• past: the answer requires information from earlier in the video. This
includes counting things that have occurred, summarizing prior activity,
comparing with an earlier moment, reasoning about causes or sequences, or
referring to a specific earlier event, object, or statement.
• future: the question sets up a trigger -- the system must wait and emit a
response only when a specified later condition occurs. Typical cues: "when X
happens, output Y", instructions to monitor and respond.
Present-tense phrasing alone does not imply current. If the correct answer needs
evidence from before this moment, it is past.
Current frame caption: "{caption}"
Question: "{question}"
Reply with exactly one word: current, past, or future.
```

Decoding configuration. Greedy decoding (temperature $= 0$, do_sample $=$ False), max new tokens $= 8$. The single-word output is parsed case-insensitively; malformed outputs default to `current`.

Caption substitution. The `{caption}` placeholder is filled with the 8-field structured caption (§B.1) of the frame nearest to the question's time stamp.

B.3 Gating Prompt

For proactive-output questions (the question sets up a future trigger of the form "when X happens, output Y"), a binary gating prompt is issued at each streaming step to decide whether the cue condition

has been satisfied at the current moment. When the gate emits "yes", a follow-up cue-response call is issued on the same clip to extract the actual cue answer; the $(\text{time}, \text{answer})$ tuple is then scored against the ground-truth event time and output string.

Gating Prompt

```
You are a gating agent watching a streaming video at time t={t_now}s.
At t={t_q}s the user asked: "{question}"
The expected output to deliver when the right moment arrives: "{gt_output}"
Captions describing what is visible:
[t={t_now}s] {caption}
Based on the captions, has the moment to deliver the expected output arrived?
Answer with only "yes" or "no":
```

Decoding configuration. Greedy decoding (temperature $= 0$, do_sample $=$ False), max new tokens $= 32$. The gate output is parsed by case-insensitive substring containment of "yes". The cue-response output is scored by case-insensitive substring containment of `gt_output`.

Caption substitution. The `{caption}` placeholder is filled with the 8-field structured caption (§B.1) of the frame nearest to the question's time stamp.

C Additional Benchmark Explanation

C.1 OVO-Bench.

OVO-Bench [27] evaluates temporal awareness in online video understanding by posing questions at specific timestamps during a video stream, rather than after the entire video has been observed. The benchmark comprises 644 videos with 2,814 human-curated question-answer pairs across 12 tasks, organized into three temporal scenarios. *Real-Time Visual Perception* tests understanding of the current scene at query time, covering Optical Character Recognition (OCR), Attribute Recognition (ACR), Action Recognition (ATR), Spatial Understanding (STU), Future Prediction (FPD), and Object Recognition (OJR). *Backward Tracing* requires the model to recall past events, with tasks including Episodic Memory (EPM), Action Sequence Identification (ASI), and Hallucination Detection (HLD). *Forward Active Responding* presents questions whose answers depend on events that have not yet occurred; the model must decide to wait rather than respond prematurely. This scenario includes Repetition Event Count (REC), Sequential Steps Recognition (SSR), and Clues Reveal Responding (CRR). Backward Tracing and Real-Time Visual Perception adopt a multiple-choice format with accuracy as the metric. Forward Active Responding employs a multiple-triggering evaluation pipeline that densely queries models along the temporal axis, using both accuracy-based and score-based metrics. The overall score is the macro average across the three scenario-level averages.

C.2 StreamingBench.

StreamingBench [26] evaluates streaming comprehension by presenting questions at different temporal positions within a video, simulating real-time user interaction during playback. The benchmark contains 900 videos with 4,500 human-curated QA pairs across 18 tasks under three dimensions. *Real-Time Visual Understanding* (10 tasks) covers Object Perception (OP), Causal Reasoning (CR), Clips Summarization (CS), Attribute Perception (ATP), Event Understanding (EU), Text-Rich Understanding (TR), Prospective Reasoning (PR), Spatial Understanding (SU), Action Perception (ACP), and Counting (CT). *Omni-Source Understanding* (4 tasks) requires integrating audio and visual signals, including Emotion Recognition (ER), Scene Understanding (SCU), Source Discrimination (SD), and Multimodal Alignment (MA). *Contextual Understanding* (4 tasks) evaluates higher-level reasoning over accumulated context, including Anomaly Context Understanding (ACU), Misleading Context Understanding (MCU), Sequential Question Answering (SQA), and Proactive Output (PO). The PO task is notable in that the model must determine the appropriate moment to respond without an explicit user query; a response is correct only if the output timestamp falls within two seconds of the ground-truth timestamp. All tasks follow a multiple-choice format with accuracy as the primary metric.

D Broader Impacts

Our work improves streaming video question answering by preserving and retrieving temporal evidence from video streams. Potential negative impacts include privacy-sensitive deployment in surveillance, workplace monitoring, or bystander recording, where persistent textual memory may expose identities, locations, or sensitive activities. Incorrect captions or temporal misrouting may also cause false alarms or misleading summaries in safety-critical settings. We recommend deployment only with user notification, access control, retention limits for the Time Dictionary, and mechanisms for deleting stored temporal records.

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